

Circuit Reference

Every circuit block that has been built and verified, with the schematic, the reasoning, the simulated result, and what each model leaves out. A dashed part is on the board but not in the netlist below it — its impedance is either folded into a lumped value elsewhere in the same circuit, or left out entirely. The checks are claims about the solid parts only.

20 blocks all checks **passing** generated from **sim/blocks/** ← Field handbook

Sensor reference →

01 Power chain

02 Sensor front-ends

03 Speed and position

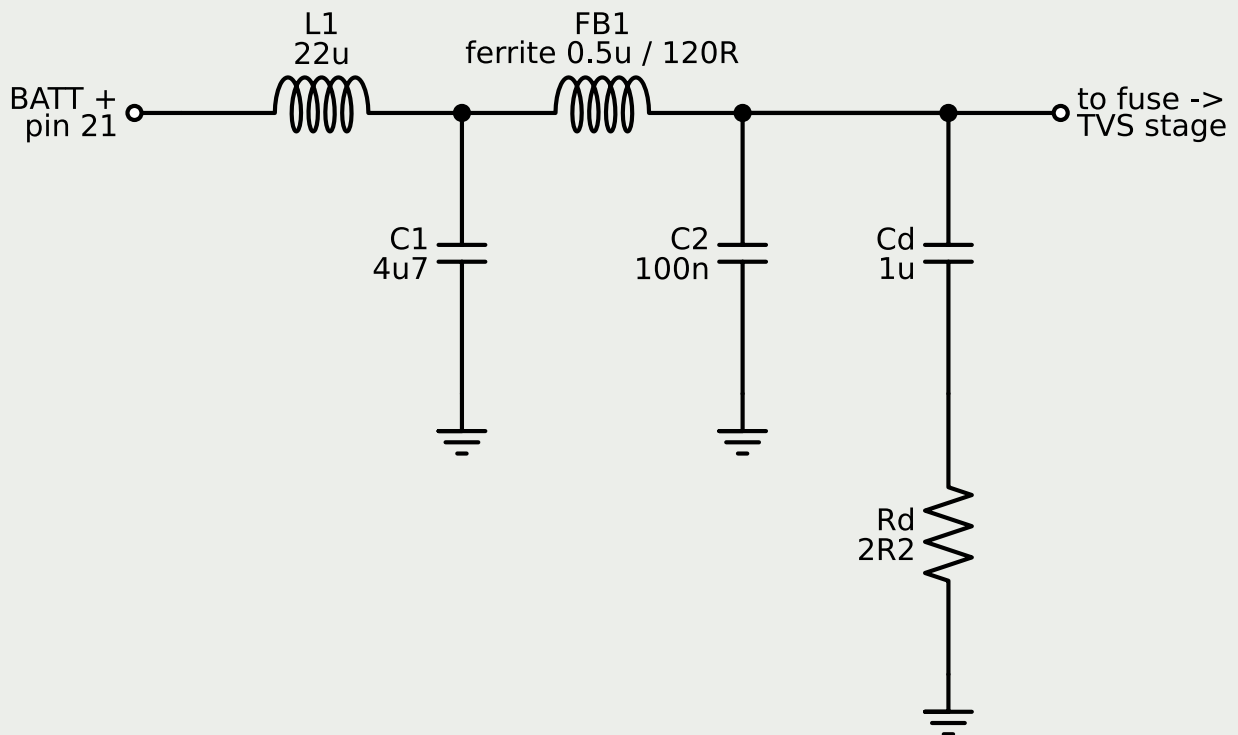
04 Injection

05 Actuators

06 Communications

01 Power chain

From the battery terminal to the MCU rail. Six stages, each one a different failure it has to survive.



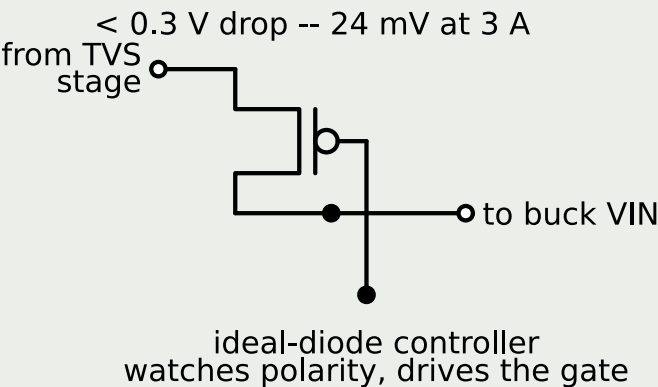
Two LC stages between the battery and everything else, measured 50 ohm to 50 ohm the way a LISN would see it.

Why it looks like that. One stage is not enough, and the reason is the capacitor: C1's own lead inductance puts its self-resonance near 1.9 MHz, and above that it stops being a capacitor. Stage two picks up there. The damper across C2 is not optional either – without it the ferrite and C2 ring at 717 kHz and hand back 12 dB right where the switching frequency's second harmonic sits.

What the model does not cover. Differential mode only.

Common-mode noise – both wires swinging together against chassis, which is most real automotive failures – needs the common-mode choke, and that cannot be modelled honestly without a layout to give it a return path.

PASS insertion loss at 150 kHz (band start)	-41.071 dB
PASS insertion loss at 400 kHz (switching f)	-51.546 dB
PASS insertion loss at 1 MHz	-74.812 dB
PASS insertion loss at 30 MHz	-141.697 dB
PASS insertion loss at 100 MHz (FM band)	-128.977 dB
PASS worst insertion loss anywhere above 150 kHz	-41.171 dB
PASS ... worst point sits at	151.5 kHz
PASS is 40 dB enough for Class 5?	unknown -- needs measured unfiltered emissions on hardware
Netlist: sim/blocks/emi_filter.cir	



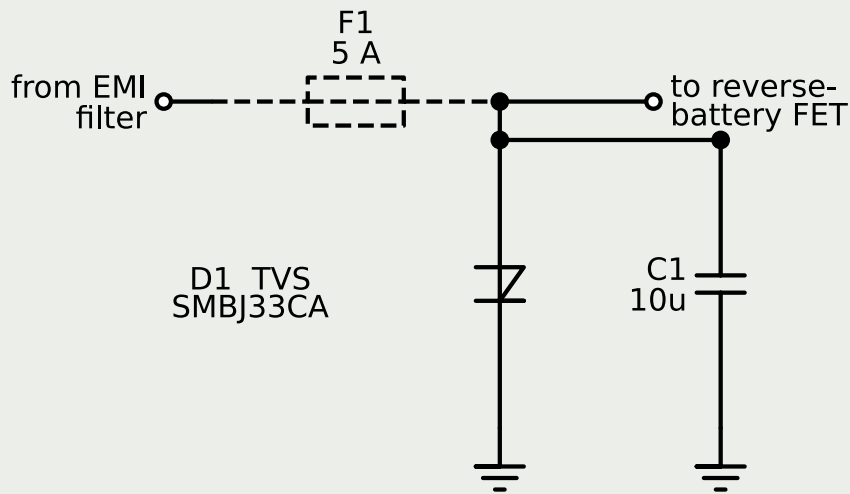
A P-FET held on by a controller that watches polarity, compared against the obvious Schottky.

Why it looks like that. Somebody will reverse the jump leads eventually. A Schottky survives it and charges you 0.46 V on every amp forever; the ideal-diode arrangement costs a controller IC and drops 24 mV. At 3 A that is 1.4 W of heat that simply never happens.

What the model does not cover. Steady state only. A real controller takes about a microsecond to turn the FET off, and current flows backwards through the body diode during that window. The switch model here turns off instantly, so it will always look perfect in reverse — that number is a datasheet question.

PASS Schottky drop at 3 A	0.465 V
PASS ideal-diode FET drop at 3 A	0.024 V
PASS FET drop stays under 0.3 V spec to 5 A	0.040 V
PASS ... Schottky does NOT meet it	0.52 V at 5 A -- fails the 0.3 V spec
PASS power burnt in the Schottky at 3 A	1.395 W
PASS ... and in the FET instead	0.072 W
PASS Schottky blocks -14 V (leak into 1k)	1.000 mV
PASS ideal-diode FET blocks -14 V	0.014 mV

Netlist: sim/blocks/reverse_battery.cir



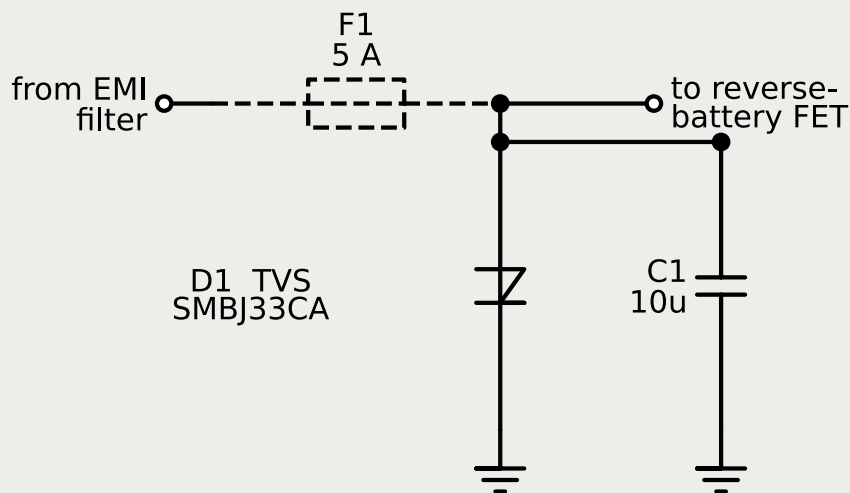
F1 dashed: on the board per spec sec.4, but this netlist folds all series impedance into Rsrc -- F1 is not modelled, and its survival under pulse 2a's repeated ~38 A stress is untested

A bidirectional TVS behind a 5 A fuse, against ISO 7637-2 pulse 2a: +112 V through 2 ohms for 50 us. Second and third stages of spec sec.4's pin 21 order — EMI filter, then fuse, then this TVS, then the reverse-battery FET.

Why it looks like that. Clamps to 50 V, which leaves 2x margin inside the buck controller's 100 V rating and keeps 60 V-class parts viable everywhere downstream.

What the model does not cover. It misses the 42 V figure the earlier documents asked for, and that figure should be dropped: it was arbitrary, with no standard behind it. The real constraints are the parts' ratings. The fuse (dashed in the schematic) is not modelled either — this netlist folds all series impedance into Rsrc — and its survival under pulse 2a's repeated ~38 A instantaneous stress has not been tested, unlike load_dump, which did check this for its own pulse.

PASS pulse actually applied (source peak)	125.500
PASS clamped voltage at buck input	49.998
PASS ... inside LM5164's 100 V rating	2.000 x
PASS ... 60 V-class parts still viable	49.998
PASS ... but misses the inherited 42 V target	50.0 V -- target was arbitrary
PASS recovers to nominal after pulse	12.981 V
Netlist: sim/blocks/transient_clamp.cir	



pulse 5b: $U_s^*=40\text{ V}$, $0.5R$, 400 ms
 D1 absorbs $\sim 46\text{ J}$ at $\sim 116\text{ W}$ -- FAILS an SMBJ-class part's rating
 F1 dashed (drawn, not in this netlist) does not rescue it either:
 fault current is 3.04 A , $0.6\times$ F1's rating, and won't clear in 400 ms

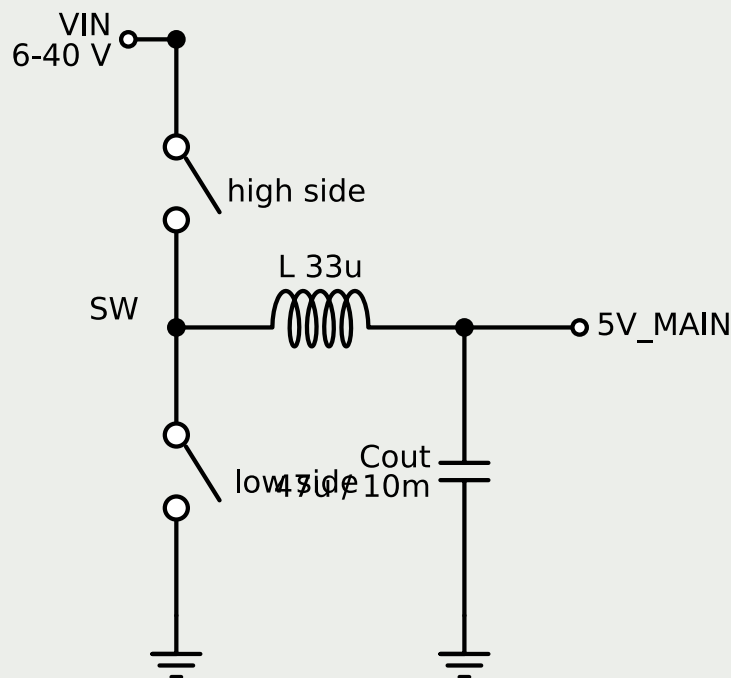
The same TVS and capacitor as transient_clamp, against ISO 7637-2
 pulse 5b: a suppressed load dump, 40 V through 0.5 ohms for 400 ms —
 four orders of magnitude longer than pulse 2a.

Why it looks like that. Clamps to 38 V , comfortably inside every voltage rating in the chain — the same story as pulse 2a. But 400 ms is long enough that the limit stops being clamp voltage and becomes dissipated energy: the TVS absorbs roughly 46 J at $\sim 116\text{ W}$ average, which is on the order of $100\times$ an SMBJ-class part's single-pulse energy rating and its continuous power rating both. This block FAILS on purpose — a TVS sized against pulse 2a's peak voltage was never sized against pulse 5b's energy, because nothing before this block asked the energy question. But the whole failure turns on one inequality: re-run at $U_s^* = 35\text{ V}$, the figure usually quoted for a 12 V suppressed dump, and the TVS never reaches its 36.7 V breakdown — 0.000 A , 0 J , no problem at all. So the finding is not that an SMBJ33CA cannot survive a load dump. It is that a TVS whose standoff sits BELOW the dump level gets forced to conduct during a normal event and then has to absorb it. The fix is a higher standoff (SMBJ40CA/45CA class), not a bigger energy rating: the TVS then stays off and the buck, rated 100 V , rides 40 V without complaint. Energy absorption only becomes unavoidable under pulse 5a.

What the model does not cover. Whether this pulse even applies: 5b assumes the alternator's rectifier is suppressed/clamping. That has not been confirmed for this genset — if it is not, the real event is pulse 5a (65–87 V, unclamped), a harder problem this block does not simulate. Also not modelled: the real pulse's exponential-decay shape (approximated here as a trapezoid, which is pessimistic in the right direction), and the TVS's actual junction thermal transient. The 5 A fuse (dashed in the schematic, not a netlist element) does not rescue this failing case either: the fault current is 3.04 A, 0.6x the fuse's rating, and it will not clear in 400 ms.

PASS pulse actually applied (source peak)	40.000 V
PASS clamped voltage through the 400 ms plateau	38.097 V
PASS ... inside LM5164's 100 V rating	2.625 x
PASS ... 60 V-class parts still viable	38.097 V
PASS TVS current through the plateau	3.044 A
PASS average power dissipated in the TVS	115.954 W
PASS total energy absorbed by the TVS over the pulse	46.440 J
FAIL energy stays within a single-pulse (~0.5 J) rating	46.440 J
FAIL average power stays within continuous (~3 W) rating	115.954 W
PASS ... exceeds the continuous rating by	38.651 x
PASS recovers to nominal after pulse	13.366 V
Netlist: sim/blocks/load_dump.cir	

400 kHz -- above CISPR 25's 150 kHz band start

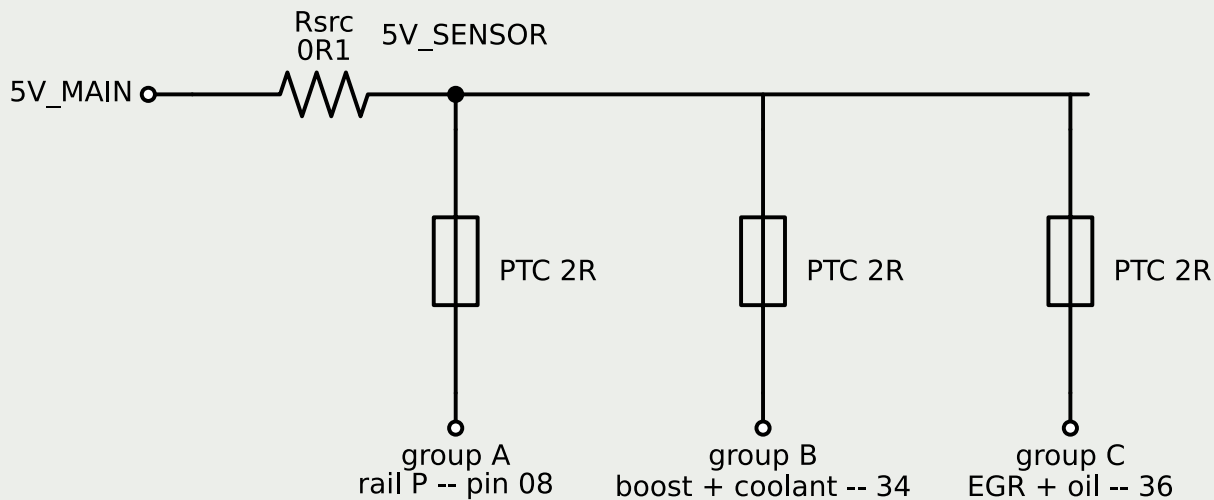


6-40 V down to 5 V at 400 kHz, run open-loop at fixed duty so the output filter is visible.

Why it looks like that. 400 kHz is chosen against CISPR 25, not for size: the conducted band starts at 150 kHz, so a fundamental below it would put every harmonic inside the measurement from the first one up. The inductor is sized at HIGH line, not low – ripple current grows with input voltage even though duty cycle falls.

What the model does not cover. Ideal switches with a fixed on-resistance. Good enough for ripple, which is set by L and C; useless for efficiency, which is set by switching overlap, gate charge and reverse recovery, none of which this model has.

PASS output at 13.5 V in (open loop, D=0.370)	4.926 V
PASS output at 40 V in (open loop, D=0.125)	4.945 V
PASS inductor ripple at 13.5 V	0.256 A
PASS inductor ripple at 40 V	0.348 A
PASS ... worst case is HIGH line, not low	1.362 x
PASS worst-case ripple as fraction of 1 A load	34.840 %
PASS output ripple at 13.5 V	13.845 mV
PASS output ripple at 40 V	21.423 mV
Netlist: sim/blocks/buck_preregulator.cir	



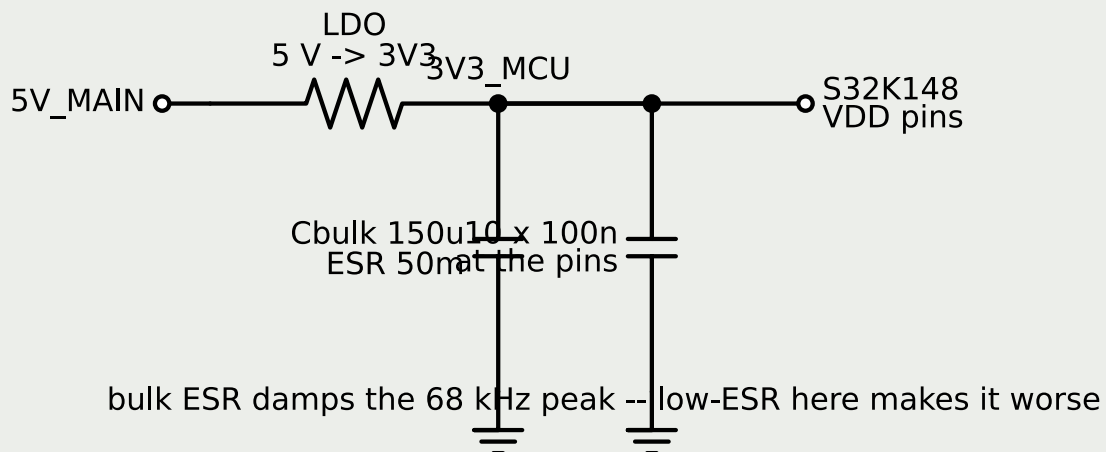
One resettable fuse per sensor group, and the same fault simulated without them.

Why it looks like that. A 5 V sensor supply shorted to engine ground is a routine fault, not an exotic one. The question is what it does to the OTHER sensors: with per-group protection the healthy groups keep 4.66 V, without it the whole rail collapses to 0.5 V and the ECU loses every analog channel at once — which means it cannot tell a wiring fault from an engine failing.

What the model does not cover. The polyfuse is modelled cold, which is the worst case for the healthy groups: the instant after the short, before the device has heated and tripped. Once tripped the healthy groups recover further. Firmware still has to notice — a tripped fuse and a dead sensor look identical from the ADC's side.

PASS healthy group with PTCs, group A shorted	4.661 V
PASS ... rail itself holds up	4.754 V
PASS healthy group WITHOUT PTCs, same fault	0.495 V
PASS ... so one harness short blinds the whole ECU	0.50 V -- every analog channel lost at once
PASS protection improves the healthy group by	9.409 x
PASS fault current drawn from the 5 V rail	2.458 A

Netlist: sim/blocks/sensor_rail.cir

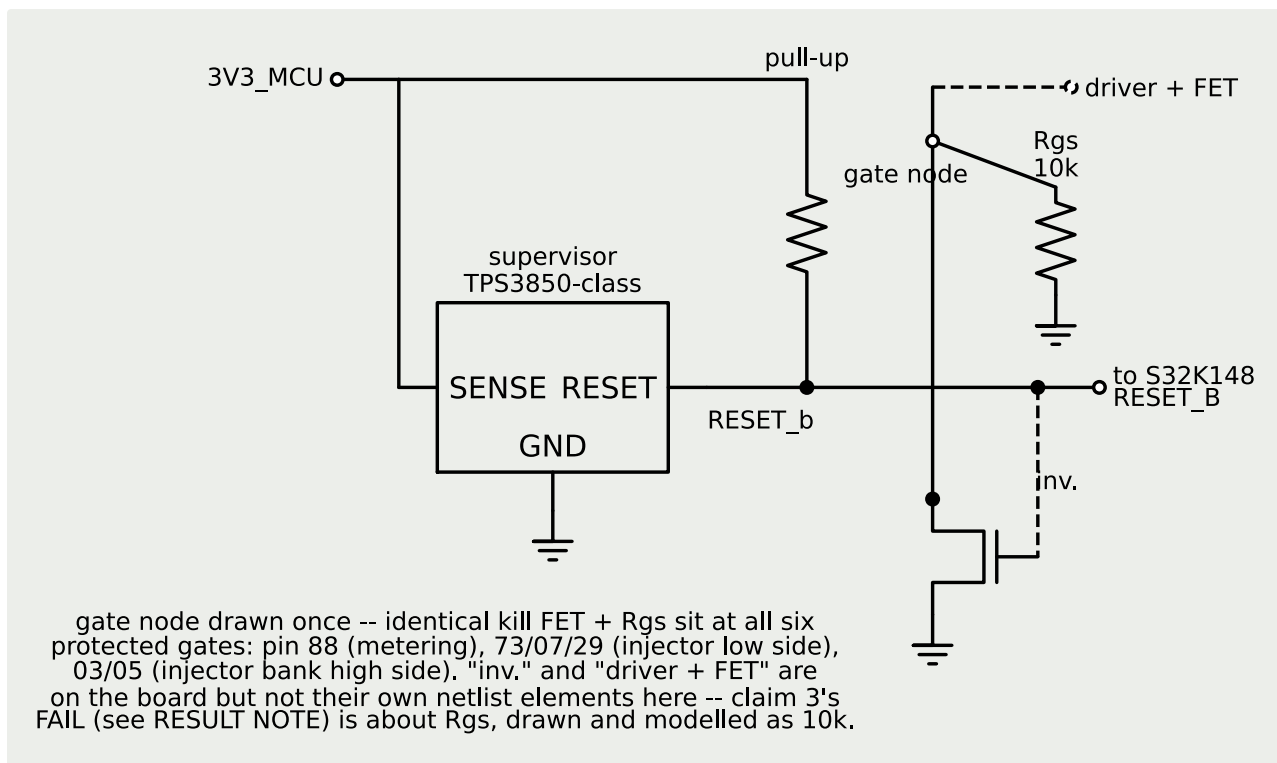


The impedance the S32K148 sees looking into its own supply, from 1 kHz to 100 MHz.

Why it looks like that. The bulk capacitor here is sized by a resonance, not by charge. The first version used 10 uF with low ESR and peaked at 3.4 ohms around 48 kHz — the regulator's own output inductance ringing against the bulk capacitance, decades away from the bulk/ceramic anti-resonance the design was drawn around. Bigger AND lossier fixes it, which is why a plain aluminium part beats a ceramic in that position. It was pulled a second time on 18 September, from 47 uF to 150 uF: at 47 uF the block cleared its target only at the nominal value of the LDO's output inductance, a number no chosen part pins down, and a 26% margin that evaporates when an unmeasured parameter is restated is not a margin. At 150 uF that resonance has dropped and damped enough that a different peak is now the tallest — the bulk/ceramic anti-resonance at 2.6 MHz, floored near 60 mOhm — and that one does not move with the LDO at all. The verdict has stopped depending on the unpinned number. More bulk beyond 150 uF buys nothing, because it does not touch the peak that now dominates.

What the model does not cover. Lumped: no planes, no via inductance, no spreading inductance, so it is optimistic above about 100 MHz — which is exactly where the sweep stops. An earlier version swept to 1 GHz and obediently reported ESL times omega at the last point as a 'peak'.

PASS peak impedance, 1 kHz to 100 MHz	60.355 m0hm
PASS ... and it has moved to the MHz anti-resonance	2.630 MHz
PASS ... i.e. bulk against ceramic, no longer regulator L against bulk	2.63 MHz -- the LDO resonance is no longer the limiter
PASS ten 100 nF parts at 100 MHz	46.923 m0hm
PASS one 1 uF part at 100 MHz	357.421 m0hm
PASS ... same capacitance, this much worse	7.617 x
PASS ten-part net at the 3.5 MHz anti-resonance	55.862 m0hm
PASS ... one-part net is LOWER there (its ESR damps it)	34 m0hm vs 56 -- more ESR, better damping
Netlist: sim/blocks/mcu_pdn.cir	



A TPS3850-class windowed supervisor watching 3V3_MCU — the same rail `mcu_pdn.cir` characterizes — whose RESET drives, through one inverting stage, a kill FET at every protected driver gate (pin 88 metering, 73/07/29 injector low side, 03/05 injector bank high side), in parallel with a standing 10 kΩ gate-source pulldown.

Why it looks like that. Research memo 11 found this board has no external watchdog, no brownout supervisor, and nothing defining output state during MCU reset — and that spec §3's other lines of defence both collapsed on 18 Sep 2026 (the fitted DSE4522 takes engine speed from this ECU over CAN, and its fuel relay signals rather than removes power), making this the first line, not defence-in-depth. Five falsifiable claims came out of that memo; two mattered most. Claim 4, the central one, PASSES with a wide margin: across six brownout ramps spanning 293 μs to 1 s, the kill path always finishes (gate below $V_{gs(th)}$, fought by an adversarial driver the whole time) before 3V3_MCU crosses 2.97 V — the S32K148's own PLL-guarantee floor — with margin that GROWS at slower ramps, so the memo's flagged-open 'slow sag' case does not threaten it. Claim 3 FAILS, and not narrowly: swept across the memo's own illustrative Cr_{ss} range under this design's own 2.8 V/μs boost-rail turn-off edge, the 10 kΩ pulldown alone lets the coupled gate voltage exceed a representative $V_{gs(th)}$ starting at $Cr_{ss} = 50$ pF — the LOW end of that range, not an extended one — and the memo's own hand arithmetic claiming '~5.6 mV' at 200 pF is off by three orders of

magnitude (560 μ A through 10 k Ω is 5.6 V). A 300 Ω –1 k Ω pulldown clears it with real margin; the fix is a component VALUE, not a new part.

What the model does not cover. The pin-mux gap is real and not papered over: which physical PTxx pins 88/73/07/29/03/05 land on is still a Phase-2 item, so their individual reset pull-state defaults cannot be stated — this model treats every protected gate generically, with no GPIO drive assumed at all except as a deliberately adversarial stimulus. The supervisor IC's own comparator propagation delay (assert side) is not sourced anywhere and is modelled as zero; claim 4's tightest margin (18 μ s, at the fastest ramp) is the one that delay would eat into first. The inverting stage and kill FET's own gate-drive dynamics are collapsed into one switch with an INFERRED Ron. Ciss=2 nF and Vgs(th)=1.0 V are class figures, not a chosen part's datasheet numbers — no injector/metering FET has been selected yet.

PASS claim 1:
gate below
Vgsth, kill
engaged
(bound: 10-40
us)

PASS ...
actual kill-
path
discharge
time

PASS claim
1b: gate
stays below
Vgsth despite
an
adversarial
driver
actively
holding it
high (GPIO-
independence)

PASS claim 2:
10k pulldown
slows the
turn-on edge
(to 9 V) by

PASS ... in
absolute
terms

PASS claim 3:
peak V(gate),
Crss=20 pF,
Rpd=10k alone

FAIL claim 3:
peak V(gate),
Crss=50 pF,
Rpd=10k alone

FAIL claim 3:
peak V(gate),
Crss=100 pF,
Rpd=10k alone

FAIL claim 3:
peak V(gate),
Crss=200 pF,
Rpd=10k alone

FAIL claim 3:
peak V(gate),
Crss=500 pF,
Rpd=10k alone

FAIL claim 3:
peak V(gate),
Crss=1000 pF,
Rpd=10k alone

PASS ... memo
11's own hand
arithmetic at ~5.6 mV -- off by 1000x ($560 \text{ uA} * 10 \text{ kOhm} = 5.6 \text{ V}$, not 5.6 mV); thi
Crss=200 pF
claimed

PASS ... so
the 10k
pulldown
fails even at
the memo's
OWN low end

PASS claim 4:
gate safe
before 2.97V
guarantee
floor,
ramp=293 us

PASS claim 4:
gate safe
before 2.97V
guarantee
floor,
ramp=585 us

PASS claim 4:
gate safe
before 2.97V
guarantee
floor,
ramp=1.17 ms

PASS claim 4:
gate safe
before 2.97V
guarantee
floor,
ramp=10 ms

PASS claim 4:
gate safe
before 2.97V
guarantee
floor,
ramp=100 ms

PASS claim 4:
gate safe
before 2.97V
guarantee
floor,
ramp=1000 ms

PASS ... tightest margin is at the FASTEST ramp tried	18 us at 293 us --
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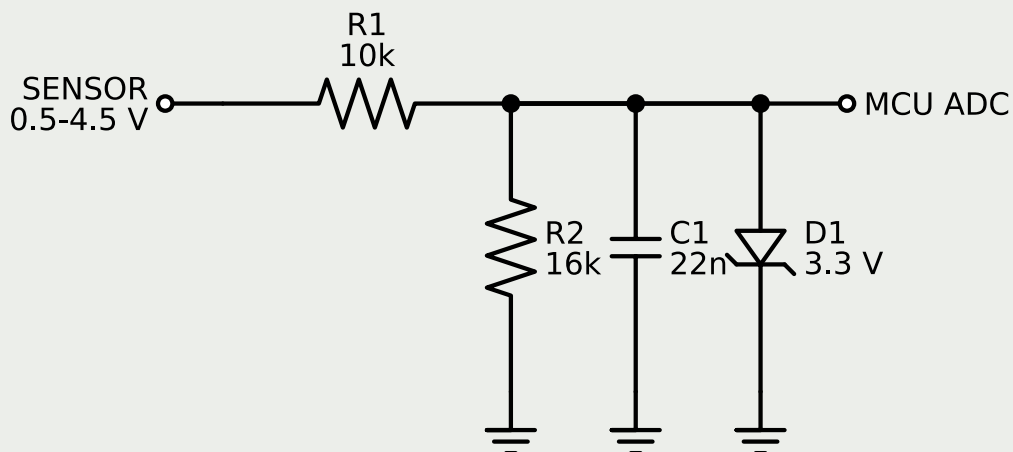
PASS claim 5: kill-path latency vs one 26.67 ms cylinder interval	
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PASS ... caveat: excludes the watchdog's own fault- DETECTION time	not
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Netlist: sim/blocks/supervisor.cir

02 **Sensor front-ends**

Everything that turns something on the engine into a number in the ADC.



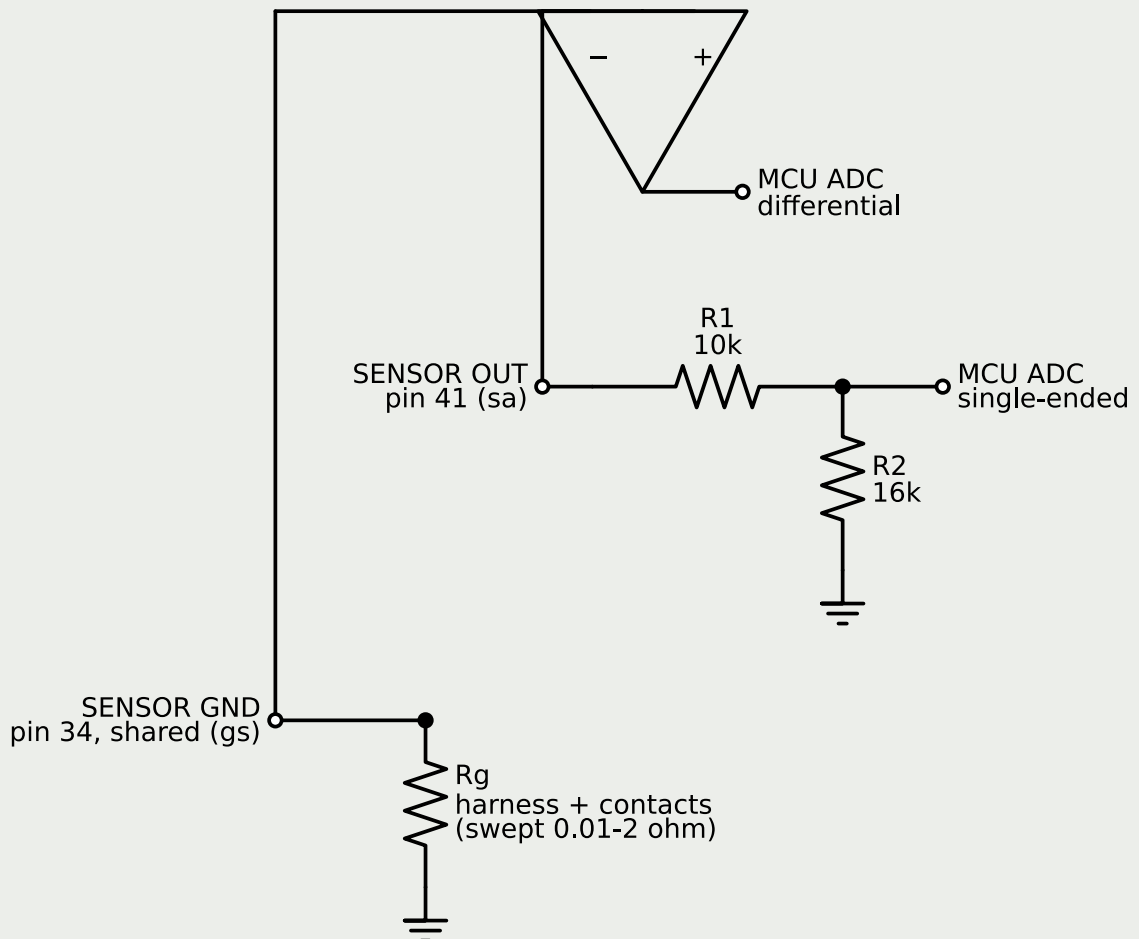
The 0.5–4.5 V pressure and position channels, scaled into a 3.3 V ADC.

Why it looks like that. Full scale deliberately lands at 2.77 V rather than hard against the ceiling, so the protection clamp can do its job on a harness fault without ever touching the signal in normal running. That costs a little span and buys a front-end 40 V cannot damage. The last four checks cover the ratiometric property itself, which this block was named for and did not model until the spec review caught it: a sensor's output is a fraction of its own supply, so a reading taken against an assumed 5.000 V carries the rail's whole error band — 10% here. Giving the rail its own ADC channel through an identical divider removes it, and the divider ratio cancels algebraically, so its tolerance drops out too.

What the model does not cover. Ground offset. Every network here measures against a perfect ground, and pins 34 and 36 are shared returns — which is why spec §4 now specifies a differential front-end for those channels and leaves this one for pin 08, the dedicated return. Also not modelled: divider tolerance matching, on which the rail correction depends, and ADC sampling time against the source impedance.

PASS sensor zero (0.5 V) maps to	0.308 V
PASS sensor full scale (4.5 V) maps to	2.769 V
PASS ... clears the 3.3 V clamp	0.531 V
PASS span used of ADC range	74.592 %
PASS low-pass corner	1,174.898 Hz
PASS loss at sensor's own 100 Hz	-0.031 dB
PASS attenuation at 100 kHz switching	-38.596 dB
PASS 40 V harness short clamps pin to	3.308
PASS ... fault current limited to	3.669 mA
PASS raw reading moves with a +/-5% rail drift	10.000 %
PASS ... so an assumed 5.000 V rail is a 10% error band	10.0% of reading, and nothing in the raw value reveals it
PASS corrected reading is flat	0.000 %
PASS correction recovers the true fraction	0.500
PASS ... divider ratio cancels, so its tolerance drops out too	measurand = V_{sig}/V_{rail} -- k appears in both and divides away
Netlist: sim/blocks/sensor_ratiometric.cir	

diff amp -- netlist's Bd: gain 16/26 (matches the single-ended ratio) plus a realistic 80 dB CMRR term



Not drawn: Ib (netlist, 20 mA into gs) -- the coolant sensor's own return current, sharing this ground via splice V3. It is what moves gs; Rg is what decides how far it moves.

The same channel measured sensor-to-sensor-ground instead of sensor-to-ECU-ground, swept against the harness return's resistance.

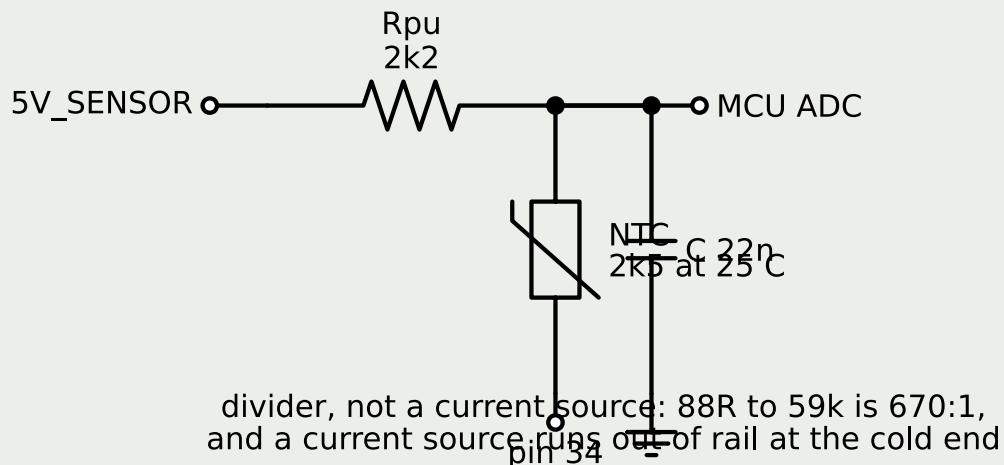
Why it looks like that. Pin 34 is spliced between the boost sensor and the coolant sensor, so it carries both sensors' return current. A single-ended input reads the resulting offset as signal. Research memo 09 found this is exactly what Deep Sea Electronics specifies against (± 2 V common mode on every sender channel) and what SEDEMAC's dedicated sensor common point exists for. Note what the OEM did with the channel that could not tolerate it: rail pressure gets pin 08, spliced with nothing. The schematic below draws one divider ($R1/R2$, the single-ended path) plus a generic amplifier for the differential path,

because that is what the netlist models — a behavioral gain block (Bd: 16/26 gain, matched to R1/R2's own ratio, plus an 80 dB CMRR term), not a second resistor divider. An earlier version of this drawing showed two dividers joined by a plain wire, which matched neither the netlist nor any circuit that would actually reject a ground offset.

What the model does not cover. The error is about half a percent of span at 1 ohm of return resistance, and that understates it. It is a bias rather than noise, so it never averages out; it grows as contacts corrode; and it is indistinguishable from a real reading — a boost pressure biased high permits more fuel, and the first symptom is smoke, not a fault code. Rg, the swept harness + connector resistance, is drawn; Ib, the 20 mA coolant-sensor current that actually moves gs, is not — it is a stimulus, not a part of this front end, and it is captioned rather than drawn for that reason.

PASS clean harness (50 mOhm), single-ended error	0.612 mV
PASS 1 ohm return, ground offset present	19.903 mV
PASS ... single-ended reading error	12.248 mV
PASS ... differential reading error	0.127 mV
PASS differential is better by	96.448 x
PASS single-ended error at 2 ohm (bad contact)	0.995 % span
PASS ... single-ended needs contact resistance below	~0.1 ohm forever, to hold 0.1% -- differential needs nothing

Netlist: sim/blocks/sensor_differential.cir

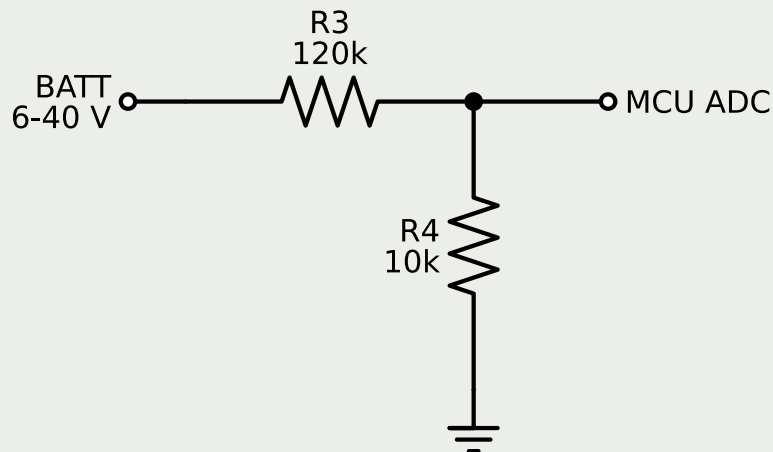


The thermistor branch of U2, as a pull-up divider and as a constant-current source, across the full -40 to 150 C range.

Why it looks like that. Memo 09 found DSE drives its temperature senders with a fixed 15 mA and measures differentially, and the obvious move is to copy it. Checked against an actual charge-air NTC it does not transfer: 88 ohms to 59 kilohms is a $670:1$ span against the $50:1$ a genset sender covers, and a current source sized to give useful signal at 150 C demands 5.9 V from a 5 V rail at -40 C. The divider's non-linearity is what saves it — it is bounded by the supply at both ends.

What the model does not cover. Which branch this engine actually needs is still U2, and one resistance measurement at the machine settles it. The self-heating figure should be revisited once the real sensor's dissipation constant is known.

PASS divider at 150 C (88 ohm)	0.192 V
PASS divider at 25 C (2.5k)	2.664 V
PASS divider at -40 C (59k)	4.820 V
PASS ... never leaves the 5 V rail	4.820 V
PASS 100 uA source needs this at -40 C	5.899 V
PASS ... but the rail is 5 V	5.9 V demanded -- out of compliance, reading clips
PASS 100 uA source at 150 C gives only	8.800 mV
PASS current that WOULD stay in compliance	84.746 uA
PASS ... leaving this at 150 C	7.458 mV
PASS divider self-heating power at 25 C	2.829 mW
PASS ... error at ~2 mW/C dissipation constant	1.414 degC
Netlist: sim/blocks/ntc_frontend.cir	



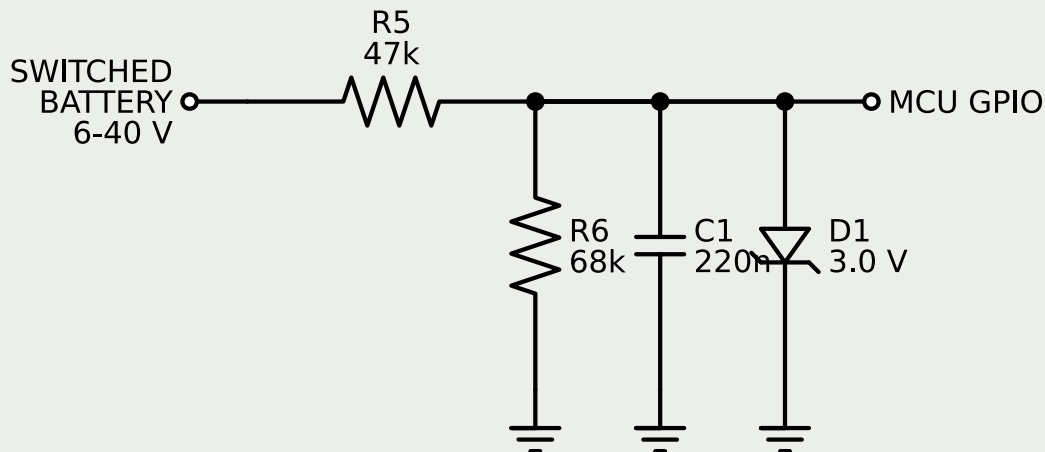
R3/R4: the adopted divider (spec sec.4, pins 04/06).
battery_sense.cir also carries a rejected R1/R2 75k/10k
divider, sized for a 5 V ADC -- not drawn here.

The 6–40 V system rail scaled into the ADC.

Why it looks like that. The inherited 75k/10k divider was sized for a 5 V ADC and reaches 4.69 V at 40 V — it would have damaged the input pin. 120k/10k peaks at 3.07 V and still resolves 573 counts at the 6 V cranking dip. The schematic below draws only the adopted 120k/10k divider, and it now carries the netlist's own designators for it, R3/R4 — spec §4 gives the same 120k/10k for pins 04/06. The rejected 75k/10k comparison is R1/R2 in battery_sense.cir; it is not drawn, since it is not the circuit on the board, and using its designators for the drawn divider (an earlier version of this drawing did) pointed a reader at the wrong resistors.

What the model does not cover. Nothing subtle. A resistive divider is exactly linear and the check confirms it, which is there to catch a wrong topology rather than a wrong value.

PASS inherited 75k/10k peak at 40 V	4.694 V
PASS ... and so exceeds a 3.3 V ADC	4.69 V > 3.3 V -- would damage input
PASS corrected 120k/10k peak at 40 V	3.069 V
PASS ... stays under the 3.3 V rail	3.069
PASS headroom below rail	0.231 V
PASS reading at 6 V cranking dip	0.462 V
PASS ... resolves to usable ADC counts	572.867 LSB
PASS divider is linear (max residual)	0.000 V
PASS slope matches 10/130	0.077
Netlist: sim/blocks/battery_sense.cir	



The switched-battery inputs, active high.

Why it looks like that. This one cannot be built as a divider at all. Reading logic high at a 6 V cranking dip needs a ratio above 0.385; staying under 3.3 V at 40 V needs it below 0.0825. No fixed ratio satisfies both, so the original was impossible rather than merely mis-valued. A 3.0 V zener sets the level and the divider only biases into it, which holds the output flat within 92 mV across the whole range.

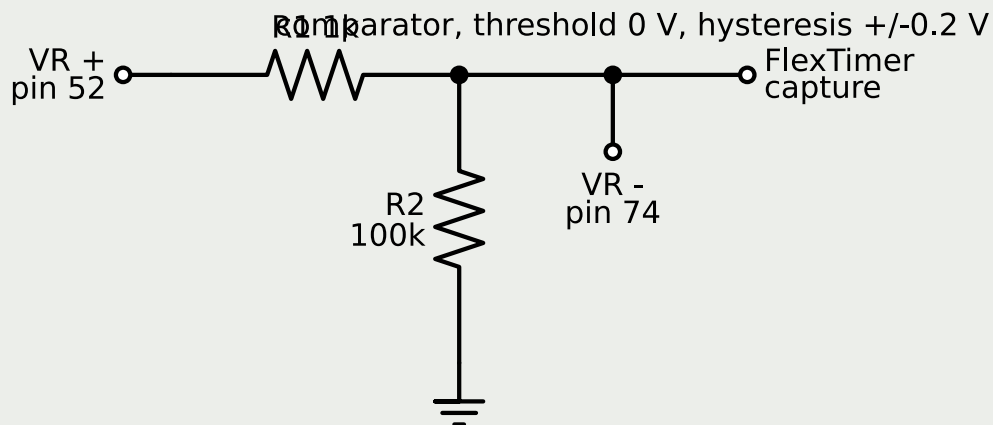
What the model does not cover. The debounce capacitor went from 100 nF to 220 nF after simulation showed the smaller value let two threshold crossings through a bouncing contact — which firmware would have read as two separate presses.

PASS never exceeds 3.6 V abs max over 6-40 V	2.954
PASS logic high at 6 V cranking dip	2.863
PASS logic high at 40 V load dump	2.954
PASS level is flat across 6-40 V (spread)	0.092 V
PASS settled high after bounce ends	2.910
PASS threshold crossings during bounce	0.000
PASS debounce time constant	6.114 ms

Netlist: sim/blocks/discrete_input.cir

03 **Speed and position**

The most safety-critical input on the board.



A variable-reluctance sensor into a zero-crossing comparator, run at both running speed and cranking speed.

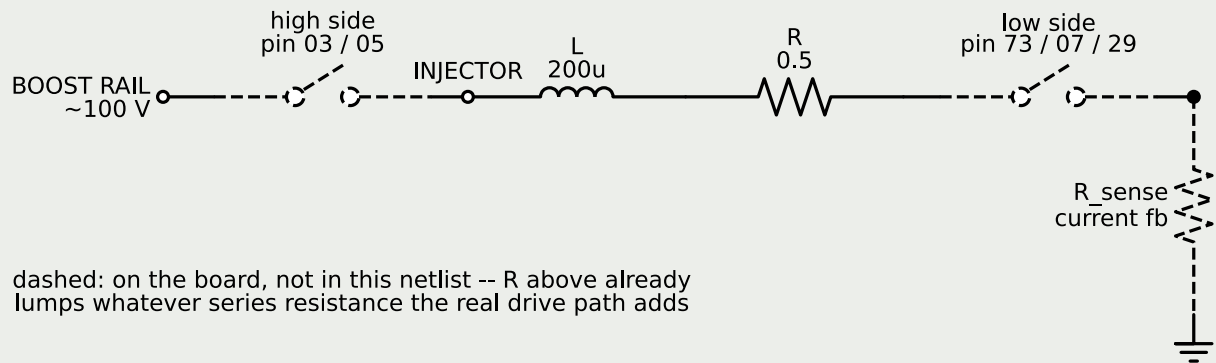
Why it looks like that. A VR sensor generates voltage by induction, so its amplitude is proportional to speed: roughly 20 V peak at 1500 rpm and 2 V at cranking. A fixed 5 V threshold, which looks perfectly sensible against a 20 V signal, produces all 30 teeth at running speed and NOTHING at cranking. That is the worst failure shape there is — it works on the bench and the engine never starts. Zero-crossing detection does not care about amplitude, and the crossing is also where the waveform is steepest and timing jitter is smallest.

What the model does not cover. A real VR signal is not a clean sine — tooth edges distort it and the missing-tooth sync gap produces one large excursion per revolution. Amplitude at true cranking speed on a cold engine with a weak battery can be under a volt, which is when starting matters most, and the hysteresis window needs checking against the real sensor rather than this 2 V estimate.

PASS zero-cross, running (1500 rpm, 30 teeth)	30.000 edges
PASS zero-cross, cranking (150 rpm, 3 teeth)	3.000 edges
PASS fixed 5 V threshold, running	30.000 edges
PASS fixed 5 V threshold, cranking	0.000 edges
PASS ... so a fixed threshold never works at 1500 rpm, blind at 150 rpm -- passes on the bench starts the engine	
PASS running tooth period	666.667 us
PASS ... jitter across the window	0.000 us
Netlist: sim/blocks/vr_conditioner.cir	

04 Injection

The most expensive stage on the board and the reason it exists.



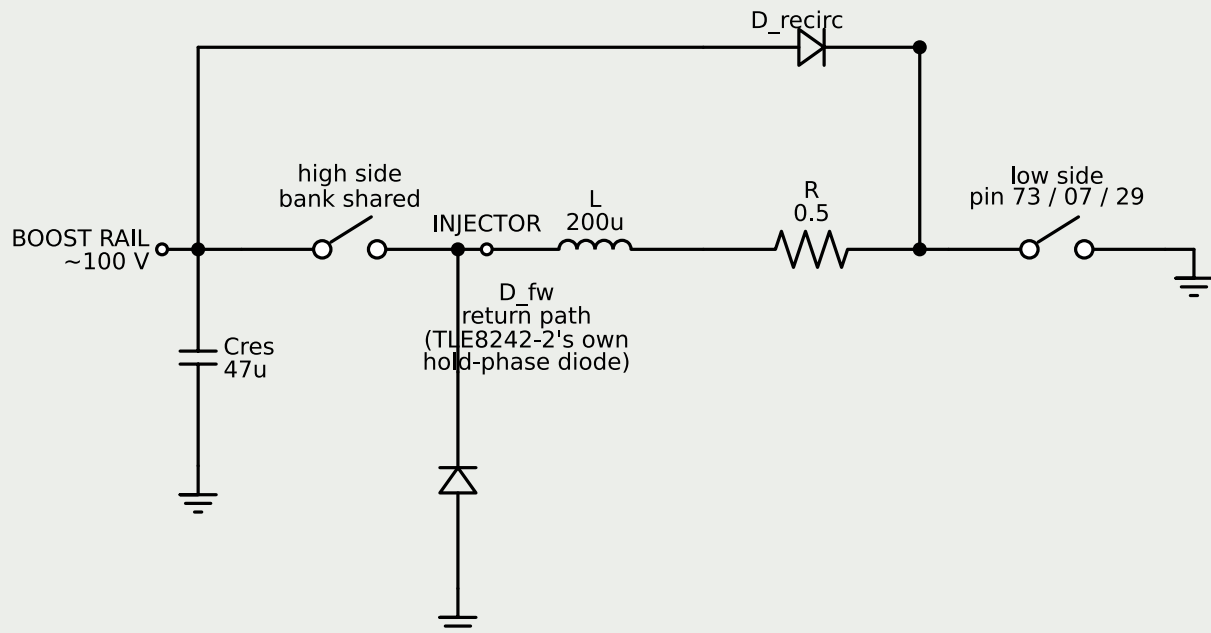
The same injector driven from the battery and from a 100 V rail, compared on time to reach the 18 A peak threshold.

Why it looks like that. 439 us from the battery, 38 us from the boost rail. On a 1 ms injection the battery drive spends 44% of the event just lifting the needle; the boost rail spends 3.8%. That ratio is the entire argument for the stage.

What the model does not cover. An RL model of the injector with no magnetic saturation and no moving needle. The envelope it sits in — 65–115 V, 12–24 A peak — is family-wide from memo 04, not this injector's measured values. That is U5. Also not modelled: the high/low-side switches and the sense resistor, dashed in the schematic — this is a turn-on-only step response straight onto L/R, and R already stands in for whatever series resistance the real drive path adds.

PASS time to 18 A peak from 13.5 V battery	439.496 us
PASS time to 18 A peak from 100 V boost	37.796 us
PASS boost is faster by	11.628 x
PASS battery drive eats this much of a 1 ms injection	43.950 %
PASS boost drive eats this much of a 1 ms injection	3.780 %

Netlist: sim/blocks/injector_boost.cir



18 A → ~0 A in ~33 us, rail bump ~+6.4 V/event (measured)
 memo 10 predicted ~36 us / ~+6.9 V ideal -- real diode and switch losses shave both, and the bump undershoots the 7.3 V on-phase droop it is paired with -- net loss, not gain

The low-side switch opening at 18 A, with a recirculation diode routing the coil's current back into the same boost cap (Cres, 47 uF) instead of a clamp that just burns it — plus a second diode (D_fw) the loop needs to close, and a comparison against freewheeling to the battery instead.

Why it looks like that. Research memo 10 surveyed real injector-driver topologies and ruled out a Zener/active clamp (Nexperia AN50003 flags repetitive active clamp as reliability-“Questionable”) and a bare battery freewheel (too slow) in favor of recirculation to the existing boost rail — confirmed production diesel-injector practice via a 1988 Marelli Autronica patent, not a textbook flourish. Simulated: 33 us to collapse 18 A to zero (memo predicted ~36 us), 25x faster than freewheeling to the battery (830 us — slower than the memo's own simplified estimate, because a real RL circuit's exponential tail matters more at 13.5 V than at 100 V). The rail bump per event is +6.4 V, under the memo's +6.9 V ideal because real diode and switch losses are modelled here, and it undershoots the 7.3 V on-phase droop it is paired with — net loss, not gain, so three back-to-back events never push the rail over its 100 V setpoint. But that verdict hinges on which current is being cut off: boost_converter.cir's own arrangement A has the battery supply HOLD current, not the boost rail, and recirculating a hold-current cutoff through the same diode has no offsetting boost-side draw to net against

— modelled separately here, it overshoots to 102 V on a single event. boost_converter.cir's charge regulation (Bchg) has no bleed path either way; it just happens not to need one for the peak-only case.

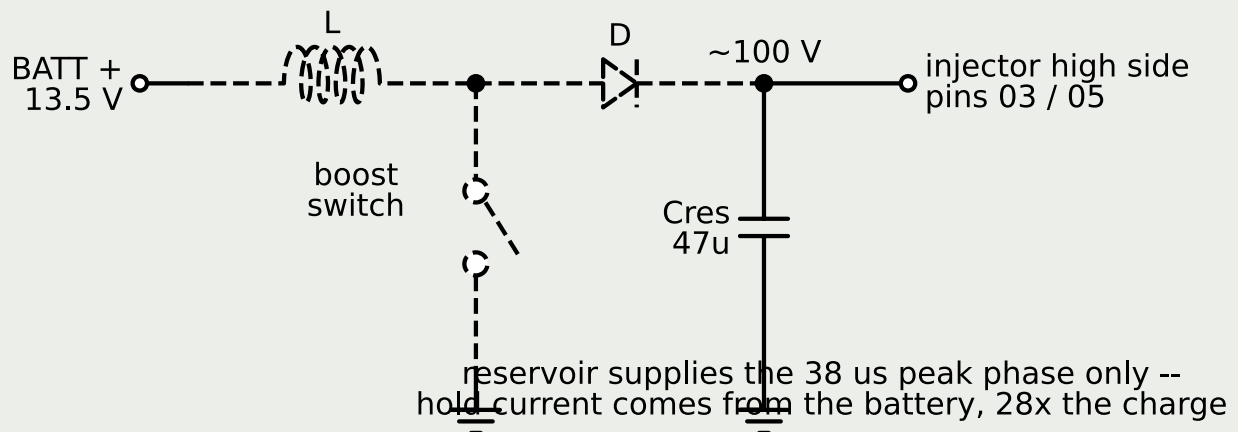
What the model does not cover. The differential-mode EMI edge (memo's own claim 5) is not modelled — the switches here are ideal with zero transition time, so any dV/dt read off this simulation would be a solver artifact, not a FET's real slew rate. Also not modelled: diode reverse recovery (Q_{rr}), the diode's thermal transient, and — inherited from injector_boost.cir — magnetic saturation and needle motion.

PASS turn-off time, 18 A → ~0 A on the boost rail	
PASS ... vs freewheel-to-battery baseline (comparison only)	
PASS ... boost recirculation is faster by	
PASS on-phase droop, pilot event	
PASS ... matches boost_converter.cir's own droop figure	
PASS recovery bump, pilot event (isolated)	
PASS ... undershoots the droop it is paired with (real losses)	
PASS rail never exceeds 100 V setpoint (3-event circuit)	
PASS rail recovered by the next cylinder (26.67 ms after post)	
PASS hold-current cutoff into the SAME cap, no offsetting draw	
PASS ... THIS is the scenario that overshoots 100 V	
PASS ... so the verdict hinges on whether hold-current cutoffs share the boost-recirculation diode	peak-only recirculation never overshoots; hold-current reci
PASS peak instantaneous power in D_recirc during turn-off	
PASS ... exceeds the ~3.6 W worst-case AVERAGE by	

PASS differential-
mode edge vs
emi_filter.cir

not modelled here (ideal switches) --

Netlist: sim/blocks/injector_turnoff.cir



The reservoir capacitor under a real injection event, with and without the hold phase drawn from the same rail.

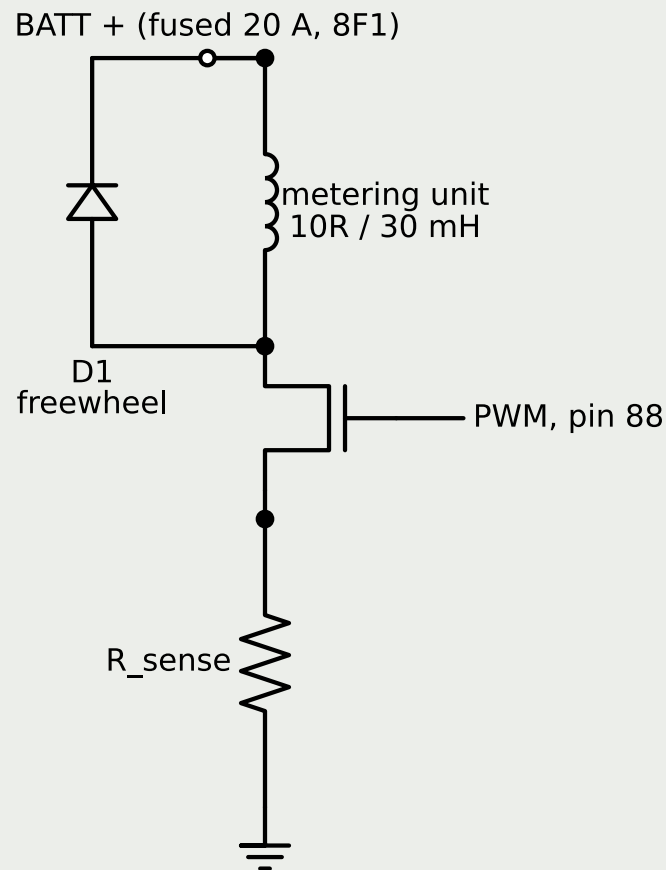
Why it looks like that. The converter cannot deliver 18 A — it delivers a trickle and the capacitor delivers the pulse, so the capacitor is the component that matters. The design question is where the HOLD current comes from, and the answer is not subtle: hold costs 29 times the peak phase's charge, so a rail supplying both collapses completely. Hold comes from the battery. 50 mA of average charging recovers the rail with room to spare in the 26.67 ms before the next cylinder fires.

What the model does not cover. Charge balance only. L, D and the switch are dashed in the schematic because the converter physically has them, but there is no inductor, no switch and no control loop in the netlist — just a current source standing in for the converter's average output. It says nothing about the converter's stability or its inrush, and the collapsed voltage in the second case should be read as 'far below usable' rather than as a number.

PASS droop, peak phase only (47 uF)	7.255 V
PASS ... rail stays usefully above battery	92.745 V
PASS droop, peak AND hold from the rail	210.913 V
PASS ... so the rail collapses	-111 V -- hold must come from the battery
PASS hold phase costs this much more charge	29.071 x
PASS recovered by the next injection (26.67 ms)	100.000 V
PASS ... 50 mA average charging is enough	100.000 V
Netlist: sim/blocks/boost_converter.cir	

05 **Actuators**

Outputs that move something.



Low-side PWM into the metering solenoid at three frequencies, all at the same duty.

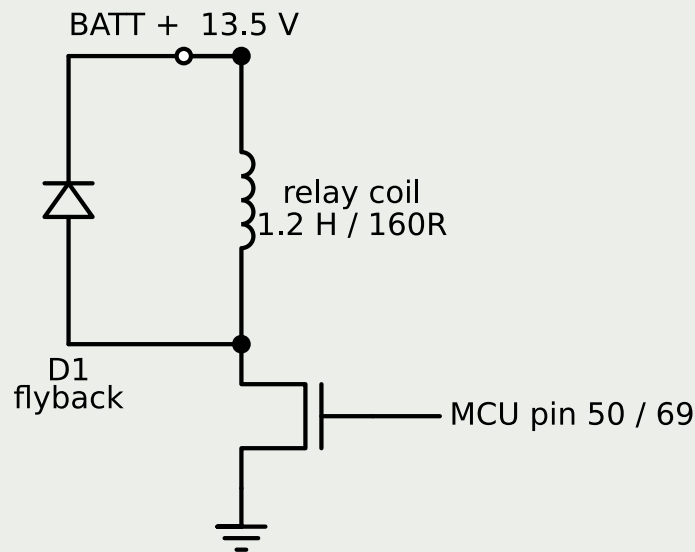
Why it looks like that. Rail pressure is controlled by the AVERAGE current through this coil, and all three frequencies produce the same average — which is what makes the control law independent of the PWM constant. What changes is ripple, and ripple is not purely a nuisance here: a proportional solenoid has stiction, and deliberate dither keeps it moving. At 100 Hz the period is comparable to the coil's own 3 ms time constant and the result is 152% ripple — that is not dither, that is chopping, and the valve follows the PWM rather than its average.

What the model does not cover. The correct dither amplitude is a property of the specific metering unit's mechanics and is manufacturer data tied to a part number — the same gap as U5, in a smaller form. Keeping the frequency a firmware constant

rather than a hardware one means it can be tuned on the machine. Coil inductance is also not constant: it falls as the armature moves and as the core saturates.

PASS mean current at 100 Hz	0.639 A
PASS mean current at 1 kHz	0.639 A
PASS mean current at 10 kHz	0.634 A
PASS ripple at 100 Hz	0.969 A
PASS ... as fraction of mean	151.786 %
PASS ripple at 1 kHz	0.118 A
PASS ripple at 10 kHz	0.012 A
PASS ripple scales as 1/f	9.974 x
PASS 100 Hz is not dither, it is chopping	152% ripple -- valve follows the PWM

Netlist: sim/blocks/metering_unit_pwm.cir



A low-side FET switching a relay coil, run with and without the flyback diode.

Why it looks like that. With the diode the drain stays at 14.3 V on turn-off. Without it the coil's stored energy has nowhere to go except through the FET.

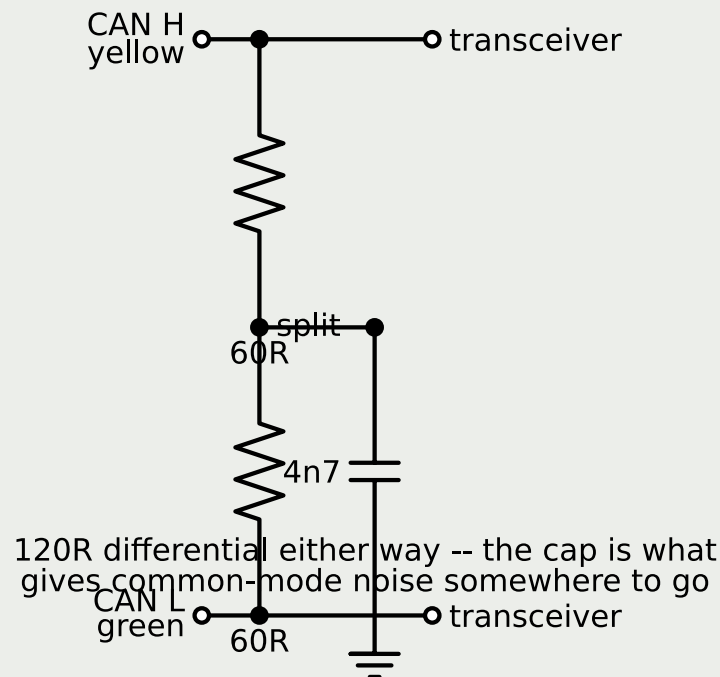
What the model does not cover. The simulated peak without the diode is hundreds of kilovolts, which is the ideal-switch model taken literally. A real MOSFET avalanches at its breakdown voltage and dissipates that energy in the die instead — once per relay operation, until it fails. The magnitude is a model artifact; the conclusion is not.

PASS drain peak WITH flyback diode	14.274 V
PASS drain peak WITHOUT diode	402,899.522 V
PASS ... diode suppresses the kick by	28,226.510 x

Netlist: sim/blocks/relay_driver.cir

06 Communications

The bus to the genset controller.



A split 60+60 termination with a capacitor to ground at the midpoint, against a single 120 ohm resistor.

Why it looks like that. Both are 120 ohms differential and a schematic label cannot tell them apart. The difference is common mode: a single resistor gives noise on both wires together nowhere to go, while the split version shunts it to ground without touching the differential signal, because the midpoint is a virtual ground for differential drive. That matters more here than on a truck. The fitted controller is a Deep Sea DSE4522, and DSE's manual is explicit that with a CAN engine the ECU transmits engine speed to it — so this bus carries a safety-relevant value, and it runs beside 415 V three-phase wiring and contactor coils in the same panel.

What the model does not cover. Termination is not the whole common-mode story; the choke is the other half and needs a layout. And this does not decide whether to fit termination at all: a 120 ohm terminator belongs at each END of the backbone, and if this ECU is not an end node, fitting one makes the bus worse. That is U4 — so the board should carry the parts with a link and let the machine decide. The DSE4522 terminates CAN on its terminals 18/19/20 and DSE specify 120 ohm cable, which settles the cable but not which nodes carry the terminators.

PASS split termination, differential at 250 kbit/s	119.993 ohm
PASS single 120R, differential at 250 kbit/s	119.993 ohm
PASS ... identical differentially	0.000 ohm
PASS split termination, common-mode at 10 MHz	30.189 ohm
PASS single 120R, common-mode at 10 MHz	500.000 kohm
PASS ... split shunts common mode better by	16,562.490 x
PASS split common-mode at the bit rate stays high	138.099 ohm
Netlist: sim/blocks/can_termination.cir	